

Expression-to-chromatin decoding of the *Arabidopsis* spaceflight transcriptome reveals a robust Polycomb-state bias in repressed genes that is not a SOG1 damage response

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Abstract

Spaceflight reprograms the plant transcriptome, but how these transcriptional changes relate to chromatin organisation is unexplored, largely because no spaceflight epigenomic assay exists for *Arabidopsis*. We built ChromDecode-At, a reproducible pipeline that decodes any *Arabidopsis* expression dataset into the chromatin-state language of the Plant Chromatin State Database (PCSD; 36 ChromHMM states) and AraENCODE (12 states), and applied it to every *Arabidopsis* RNA-seq dataset in NASA's Open Science Data Repository (OSDR; 33 accessions). Genes down-regulated by spaceflight are strongly concentrated in Polycomb-repressed chromatin: in OSD-37 seedlings, the AraENCODE Repression state has an odds ratio of 8.0 (FDR 3×10^{-105}) and PCSD state S15 (H3K27me3 + accessible DNA) one of 18.0. The direction replicates, with a smaller effect, in an independent ISS centrifuge dataset (OSD-314) once its microgravity and Mars-gravity arms are analysed separately. After sample metadata was recovered from NCBI BioSample for datasets with undocumented designs, the signal also replicates in ISS-grown roots in two ecotypes (odds 3.1–3.7, FDR $< 10^{-6}$; Col-0 in OSD-193 and OSD-218, and Wassilewskija (Ws) in OSD-218). A pre-submission audit re-derived every contrast from the OSDR study records and corrected three sample-coding errors (Fig. 7). The repressed class is a coherent peroxidase and oxidant-detoxification module (11 GO terms significant even against a state-matched Polycomb control), and all 10 of its top GO terms replicate in OSD-314. Its promoters are enriched for AT-hook (AHL) and ZHD-family motifs, an architecture that tracks the Polycomb state itself rather than the flight response. A reanalysis of the only *Arabidopsis* ChIP-seq in OSDR (SOG1-3xFLAG after bleomycin-induced DNA damage) finds no SOG1 binding at redox-module or Polycomb-down promoters, which excludes the canonical SOG1-dependent damage-repression pathway. Whole-genome bisulfite data (OSD-217) show no significant flight-induced methylation change at the module. A power analysis indicates that about 6–8 replicates per group of a single genotype are needed for a stable root-level enrichment test (9–12 when genotypes are mixed), and a stratified analysis of OSD-120 shows that its 12-sample design cannot resolve a flight $\times$ light interaction. The Polycomb-state bias of the spaceflight repression response is robust across tissues and campaigns, is not a SOG1-style damage response, and its upstream regulator remains unidentified. It is a concrete, testable target for spaceflight epigenomics.

Keywords: *Arabidopsis thaliana*; spaceflight; chromatin states; Polycomb; H3K27me3; PCSD; AraENCODE; NASA OSDR; RNA-seq; SOG1

1 Introduction

Plants are central to bioregenerative life support for space exploration, and their transcriptomes respond extensively to spaceflight [1–3]. Chromatin state, the combinatorial pattern of histone

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marks, DNA methylation and accessibility, constrains which genes a cell can express. Polycomb repression (H3K27me3) is a canonical mechanism of stable, heritable gene silencing in plants. Whether spaceflight transcriptional repression targets genes in a chromatin-state-dependent manner has never been asked, because no spaceflight chromatin assay exists for any plant. We verified that OSDR contains no spaceflight H3K27me3 ChIP-seq; the only *Arabidopsis* ChIP-seq it holds is a ground-based DNA-damage experiment (OSD-496).

ChromDecode-At sidesteps this gap. Given an expression dataset, it maps every gene to its (ground-state) chromatin context using the PCSD 36-state ChromHMM model [4, 5] and AraENCODE 12-state annotations [6]. It then tests whether differentially expressed genes are concentrated in particular chromatin classes. Here we present the full analysis programme: decoder construction and validation, a learned expression-to-state model (negative), an OSDR-wide sweep at three state resolutions, robustness diagnostics, functional characterisation of the repressed module, methylation tests, a SOG1 ChIP-seq mechanism test, a tissue-matched root replication with metadata forensics, a power analysis, and a flight $\times$ light interaction bound.

2 Results

2.1 A hybrid decoder validates on OSD-37

ChromDecode-At harmonises AGI identifiers and computes differential expression (linear models with ecotype covariates, BH-FDR [7]). It maps genes to chromatin states via TSS-anchored coordinates (gene body + 2kb upstream; 100% PCSD and AraENCODE coverage; 37.7% Plant-CADB ACR overlap [8]), and tests state enrichment of up- and down-regulated genes against the expressed genome with Fisher’s exact test (Table D1).

The first validation run used GeneLab VST counts for OSD-37 (28,067 genes $\times$ 56 samples; four ecotypes; 117 up / 53 down at padj < 0.05 , $|\log_2 \text{FC}| > 1$; Table S1). Up-regulated genes were enriched for accessible-promoter chromatin (odds 3.22, FDR 3.0×10^{-7}) and depleted from heterochromatin/TE (0.28) and Polycomb states (0.32; Table S2). Down-regulated genes were strongly enriched for Polycomb-repressed chromatin (odds 11.46, FDR 1.4×10^{-14}) and depleted from accessible promoters (0.14; Table S3; Fig. 1). The two independent state annotations agreed, and locus controls behaved correctly: *FLC* fell in a repression state, while *ACT2* and *PHYB* were active (Fig. S1).

2.2 A learned expression-to-state model fails honestly

An elastic-net multinomial model predicting state group from eight expression-distribution features reached a CV accuracy of 0.513 against a majority baseline of 0.588 (Table S4). Version 2 retrained it on tissue-matched pairs: AraENCODE seedling expression, 28,727 genes and 11 features including tissue-specificity statistics. It used nested CV and genomic-block CV (100-kb bins) and reached accuracy 0.533–0.554 against a baseline of 0.549. Transfer to leaf, root and OSD-37 spaceflight data was ≤ 0.548 in every case (Table S7, Table S9). Expression distributions do not predict chromatin state. The block CV shows the negative is not a CV artefact, and it motivates the deterministic coordinate-based backbone that carries every later analysis.

2.3 An OSDR-wide sweep: the signal replicates

The backbone ran on every *Arabidopsis* RNA-seq dataset in OSDR (33 accessions), with contrasts auto-detected from metadata and sample-name conventions (Table S10). It produced full DE + enrichment for 5 datasets and expression-only decoding for 18; 10 had no processed counts. DE was computed on $\log_2$ -transformed counts with ecotype covariates. An earlier raw-scale version inflated DE roughly fivefold and was corrected. Under the corrected model, OSD-37 [9] yielded 529 up / 526 down genes (Table S52). Re-running the sweep for all five DE datasets reproduced every count and every 12- and 36-state odds ratio exactly (Table S50). It also regenerated

the corrected 5-group enrichment (Table S51): OSD-37 down-regulated genes are enriched for Polycomb_repressed (odds 7.01, FDR 1.7×10^{-92}) and depleted from Heterochromatin_TE (0.07) and Accessible_promoter (0.38).

At AraENCODE 12-state resolution (Table S11; Fig. 2), down-regulated genes were enriched for the Repression (Polycomb) state in OSD-37 (odds 8.0, FDR 3×10^{-105}). OSD-314 [10] is treated separately below because the sweep mis-coded its design. OSD-37 up-regulated genes were enriched for Active (1.57) and Transcription2 (5.4) and almost absent from Heterochromatin (0.03). At PCSD 36-state resolution (Table S12), OSD-37 down-regulated genes converged on the H3K27me3/accessible-promoter block. Odds were 18.0 for S15 (H3K27me3 + accessible DNA), 6.4 for S16, 3.9 for S14 and 3.3 for S13 (all FDR < 0.001). Of the 526 OSD-37 down-regulated genes, 265 sit in Polycomb states S11–S15. Constitutive heterochromatin (S29–S36) was depleted in both directions across datasets. Spaceflight-repressed genes are therefore a coherent class of promoter-accessible, Polycomb-marked euchromatic genes. One derived claim from the uncorrected DE (OSD-314 down genes in Active chromatin) is retracted.

2.4 OSD-314: the direction replicates, the magnitude does not

OSD-314 is a three-arm ISS centrifuge experiment: 0 g ($n = 5$), 1 g ($n = 6$) and Mars 0.3 g ($n = 6$), crossed with red light or darkness (Table S53). The sweep’s sample-name parser recognised the 0G and 1G tokens but not “03g”. The contrast-coding step then treated the six unparsed Mars-gravity samples as treatment, so the published contrast was 0 g + 0.3 g pooled (11) against 1 g (6), with no light term. Coding gravity and light explicitly (Table S54) gives far more DEGs: 1,457 up / 1,921 down for 0 g vs 1 g, and 1,105 / 1,238 for 0.3 g vs 1 g, both with light as a covariate (Table S55–Table S56). The Polycomb bias of down-regulated genes keeps its direction at a much smaller magnitude. AraENCODE Repression odds are 1.38 (FDR 2.7×10^{-6}) for 0 g and 2.04 (FDR 2.6×10^{-21}) for 0.3 g. The 5-group Polycomb odds are 1.23 and 1.76 (Fig. 7a). All 0 g sample identifiers share a different prefix from the 1 g and 0.3 g samples, which suggests a separate processing batch. We could not confirm this from the study record, so the 0 g contrast may be batch-confounded.

Up-regulated genes. In the published (pooled) contrast, OSD-314’s 47 up-regulated genes were strongly enriched for Polycomb CDS states: S11 (odds 22.4), S12 (4.2) and S13 (5.7) (Table S12). Under correct coding, up-regulated genes remain enriched for Polycomb chromatin (5-group odds 2.59 and 2.76, FDR $< 10^{-40}$) and for S11 (odds 2.05, FDR 4.5×10^{-5} ; and 1.95, FDR 6.8×10^{-4}). The “escape from Polycomb repression” reading therefore survives in direction, but at about a tenth of the published S11 effect (Fig. 7b).

Diagnostics on the corrected contrasts. The five robustness diagnostics were first run on the published 47-gene set (Fig. S21; Table S13–Table S17). We re-ran them with the same code and random seed on both corrected contrasts (Table S58–Table S62). Re-running the published contrast reproduced Table S13, Table S15 and Table S16 exactly (Table S64), so the code paths are unchanged (Fig. 3). The diagnostics keep that script’s odds definition, which counts DEGs that have no PCSD state as outside every state. Their odds are therefore slightly lower than the sweep-style values above (e.g. S11 1.90 vs 1.95 for 0.3 g). (i) *Null calibration.* In both corrected contrasts, every tested Polycomb state exceeds the 99th percentile of 10,000 random gene sets of equal size (empirical $p \leq 4 \times 10^{-4}$). For 0.3 g vs 1 g ($n = 1,105$ up genes), the odds are 1.90 for S11, 1.84 for S12, 2.95 for S13 and 3.61 for S15. For 0 g vs 1 g ($n = 1,457$) they are 1.83, 1.55, 2.91 and 3.09. The enrichment has therefore moved: the published S11 concentration (22.4) does not survive, while S13 and S15, which were null in the published set (S15 odds 1.4, empirical $p = 0.51$), carry the corrected signal. (ii) *Downsampling control.* This test asks whether a small gene set could reach the observed odds by chance. It is not applicable to the corrected contrasts: they have more up-regulated genes (1,105 and 1,457) than the 529 OSD-37 genes it draws from,

and at these set sizes small-sample inflation is no longer a concern. (iii) *Threshold grid*. At padj < 0.05 , S11 odds rise with DE stringency from 1.03 to 1.90 to 3.16 (0.3 g) and from 0.99 to 1.83 to 3.55 (0 g) as the $|\log_2 \text{FC}|$ threshold increases from 0.5 to 1 to 2 (Table S60). (iv) *Threshold-free enrichment*. The 1,944 Polycomb genes give a running-sum maximum of 0.196 (0.3 g) and 0.163 (0 g), and no permutation of 1,000 reached it in either contrast ($p < 0.001$; Table S61). The curves also dip below zero at the down-regulated end (minimum -0.135 and -0.091), so Polycomb genes are over-represented at both extremes of the ranking. (v) *Gene identity*. 255 (0.3 g) and 302 (0 g) up-regulated genes sit in these Polycomb states (Table S62). In summary, the “escape from Polycomb repression” reading survives correct coding and every applicable diagnostic, but as a broad, moderate enrichment (odds 1.5–3.6) across S11–S15 rather than the 22-fold S11 effect originally reported.

2.5 The repressed class is a redox/peroxidase module in AHL/ZHD promoter architecture

We compared three gene sets: the target (Polycomb-down, $n = 265$), control 1 (non-Polycomb-down, $n = 241$) and control 2 (state-matched Polycomb non-DE, $n = 3,303$).

GO. Against the genome, 18 terms were significant (TAIR GAF, hypergeometric, BH-FDR [11]; Table S19), dominated by oxidative stress. They include peroxidase activity ($11.1\times$), heme binding ($7.2\times$), hydrogen peroxide catabolic process ($13.6\times$) and cellular oxidant detoxification ($8.8\times$). Eleven terms remain significant against the state-matched Polycomb control, so the flight-repressed subset is functionally a redox module, not a random Polycomb slice (Fig. 4).

Motifs. We scanned 897 JASPAR 2024 plant motifs [12] over strand-aware 2 kb promoters with per-motif $p < 10^{-4}$ thresholds (Table S18). Against the genome, 268 motifs were significant. They include ZHD1 ($2.9\times$), the AT-hook family AHL12/13/25 (up to $5.7\times$) and NTL9. No motif survives BH-FDR against the state-matched control (best nominal: NTL9, odds 2.1, padj 0.084), so the motif layer tracks the Polycomb state, not the response. Promoter GC content is similar (target 0.310 vs control 2 0.320; Table S20), which excludes GC artefacts. The published OSD-314 replication used the pooled contrast and only 18 Polycomb-down genes (Table S21). Repeated on the corrected contrasts (Table S63; Fig. 3d), the module replicates with far more power. Of the top 10 OSD-37 GO terms, all 10 are enriched at BH < 0.05 among the 268 Polycomb-down genes of the 0.3 g contrast, and 9 of 10 among the 314 of the 0 g contrast. Folds are $4.2\text{--}8.4\times$ for the redox terms (0.3 g), for example monooxygenase activity $8.4\times$, peroxidase activity $5.3\times$ and heme binding $6.8\times$. The repressed redox module is therefore a replicated feature of an independent ISS experiment, not just of OSD-37.

2.6 No flight-induced methylation change at the module

With no spaceflight H3K27me3 data, we tested the slow-turning epigenetic layer using OSD-217: Wassilewskija (Ws) ecotype, root and leaf WGBS, flight vs ground [13]. These data were processed by the authors against the ecotype-matched Ws reference assembly with bsmapp and MOABS [13–15]. Gene-body tests found no significant change for the 23-gene redox module (minimum padj 0.07; root CHG nominal $p = 0.012$; Table S22). Promoter-window tests (2 kb, 100-bp bins, 21,424 genes) also found nothing that survives BH-FDR (minimum padj 0.099; Table S23–Table S24). The module shows directionally consistent nominal CHG gains in both tissues ($p \approx 0.011$). Baseline promoter CG methylation of the Ws-defined Polycomb-down module is elevated in root (median 0.52 vs 0.02, $p = 0.043$, $n = 6$; not BH-significant). Flight $\log_2 \text{FC}$ correlates only weakly between OSD-217 (Ws root) and OSD-37 (Col-0 seedlings): $r = -0.17$ over 63 genes (Table S25). This reflects ecotype, tissue and campaign confounding, and motivates the tissue-matched replication below.

2.7 SOG1 does not bind the module: excluding the damage-response pathway

SOG1 is the plant NAC-family master regulator of the DNA-damage response [16]. If spaceflight repression of the redox module were a SOG1-style damage response, SOG1 should occupy the module’s promoters. We reanalysed OSD-496 [17]: SOG1-3xFLAG seedlings treated with bleomycin, six single-end libraries of 13.7–23.4M reads. We aligned the reads to TAIR10 with HISAT2 [18] (`-no-spliced-alignment`) and called MACS2 [19] narrow peaks ($q < 0.01$, IP vs matched-timepoint input). This gave 56 peaks at 20 min and 100 at 1 h. SOG1-IP vs wild-type-IP gave 160/180 peaks. Of the 1-h peaks, 67/100 overlap a 2 kb promoter, and the median peak-to-TSS distance is 194 bp.

The result is a clean negative (Table S26; Fig. 5). Across the peak union, no SOG1 peak falls in a redox-module promoter (0/23) or a Polycomb-down promoter (0/265) at either timepoint (all Fisher padj = 1.0). Three of 3,303 background promoters are bound. The spaceflight repression of the module is therefore not a recapitulation of SOG1’s direct damage targets.

2.8 Tissue-matched root replication, with metadata forensics

We enumerated all 62 *Arabidopsis* OSDR studies (33 with RNA-seq, 10 mentioning root tissue) and resolved the field by QC (Table S27). Each candidate’s platform and genotypes were then re-read from its OSDR study record and BioSample titles during the pre-submission audit. ISS root datasets: OSD-120 (CARA root tips [3]), and OSD-193 [20] and OSD-218, both from the APEX-03-2 payload. OSD-193 and OSD-218 ship counts keyed by GSM/SAMN accessions with empty OSDR factor values, so every sample ID was resolved to its condition via NCBI BioSample (e.g. “Spaceflight, 8 days old, Col-0 Rep4”; Tables D4–D7). OSD-193 is Col-0 and *sku6*; OSD-218 is Col-0 and WS (16 samples each; OSDR writes the Wassilewskija ecotype as both “Ws” and “WS”). Suborbital datasets: OSD-624, Col-0 on Virgin Galactic Unity 22 [21]; and OSD-406, Col-0, WS and *sku5* on Virgin Galactic VP-03 and Blue Origin NS-12 [22]. OSD-281 was excluded on recovery because it holds 16 *sku5*-mutant and 16 Ws samples and no Col-0. Documented exclusions are OSD-217 (Ws), OSD-219/427 (no counts), OSD-411/480 (EMCS partial gravity) and OSD-208 (no flight factor).

ISS roots. With age as a covariate, OSD-193 Col-0 WT (8 v 8) gave 97 up / 181 down, Polycomb odds 3.06, padj 2.2×10^{-10} (Table S29, Table S31–Table S32). The v8 run of OSD-218 entered all 32 samples as “Col-0” because the BioSample parser did not recognise “WS”. Re-run by genotype (Table S48), Col-0 alone (8 v 8) gives 96 up / 182 down, odds 3.12, padj 7.6×10^{-11} (Table S49), and WS alone (8 v 8) gives 81 / 82, odds 3.74, padj 6.5×10^{-7} . The pooled run with genotype as a covariate gives odds 3.06. The v8 pooled estimate (odds 3.14, padj 4.7×10^{-7} ; Table S33–Table S34) is therefore numerically unchanged, but it is now attributed correctly: the root-level Polycomb bias replicates in *two ecotypes*. OSD-120 restricted to Col-0 WT collapses to 2 down genes after light/day covariate control (Table S38–Table S39); its power is examined below.

Suborbital roots. OSD-624 (minutes of microgravity on a suborbital flight) is null: 36 down genes, odds 0.76, padj 0.81 (Table S35–Table S37). The v8 analysis treated OSD-624 as an orbital root dataset and read its null as a campaign difference. The study record shows it is suborbital, so the null no longer weighs against the ISS replications. The v8 run of OSD-406 pooled three genotypes and two rockets with no covariates and found 1 down gene (Table S40–Table S41). With genotype and rocket as covariates it gives 23 up / 38 down, Polycomb odds 3.17, padj 0.0074 (Table S48). Genotype and rocket are partly confounded in that design (*sku5* flew only on Blue Origin, Col-0 only on Virgin Galactic), so we treat this as a discovery-level observation. The Col-0 subset alone (3 v 3) has no DEGs.

In summary, the Polycomb enrichment of flight-repressed genes replicates in every adequately powered ISS root contrast (OSD-193 Col-0; OSD-218 Col-0 and WS) at roughly half the seedling

effect size. The corrected 5-group Polycomb odds for OSD-37 is 7.01 (Table S51). OSD-193 and OSD-218 share the APEX-03-2 payload, so they are not fully independent replications.

2.9 Power: replicates needed for a stable root-level enrichment test

How many replicates would a root study need? Effect sizes were anchored to this project’s own observations (odds 3.1 at root level, 7.0 at seedling level) with a background Polycomb fraction of 0.141 (4,812 of 34,054 genes), $\alpha = 0.05$ and 80% power. Simulating the enrichment test alone shows that ~ 40 down-regulated DEGs are needed at root-level odds, or ~ 15 at seedling-level odds (Fig. 6b; Table S42). Running the full pipeline on k replicates per group subsampled from OSD-218 (20 draws each; Table S43), power is 0–15% at $k = 2$ –6, 50% at $k = 8$, and 100% at $k = 12$ –16 in a clean design. At $k = 16$ the draws reproduce the full result exactly (117 down genes), which validates the pipeline. Interpolating gives ~ 9 –10 replicates per group in a clean design and ~ 11 –12 with OSD-120’s light $\times$ day covariate burden. OSD-120’s Col-0 WT contrast (6 v 6) sits at ~ 10 –15% power on this pooled curve. The subsampling pool mixes the Col-0 and WS samples of OSD-218 without a genotype term, which adds draw-to-draw composition noise.

Col-0-only power. Repeating the subsampling on the 16 Col-0 samples alone (8 v 8; Table S65–Table S66; Fig. 8a) gives power of 0–25% at $k = 2$ –4 and 70% at $k = 6$ (mean 84 down genes). At $k = 8$ every draw is the full Col-0 set, which reproduces the Col-0-only result exactly (182 down genes; Table S49) and is significant, but this is a single result rather than a sampling estimate. Before running it, the script reproduced Table S42–Table S43 exactly (Table S68). With two nuisance covariates mimicking OSD-120’s burden, Col-0 power is 25% at $k = 6$. A genotype-homogeneous design therefore needs about 6–8 replicates per group, and the pooled estimate of 9–12 is an upper bound inflated by genotype mixing. The Col-0 pool is too small to resolve the curve above $k = 6$, and draws at $k = 6$ overlap heavily, which may flatter that point. OSD-120’s Col-0 WT contrast (6 v 6, with a light covariate) sits between the 25% and 70% curves.

2.10 Bounding the flight $\times$ light interaction in OSD-120

The additive model’s collapse could hide a real flight $\times$ light interaction absorbed into the light term. The Col-0 WT design is a balanced 2×2 factorial (FLT/GC $\times$ Alight/dark, 3 replicates per cell, all Day 13), so the interaction is directly estimable (Table S44–Table S47; Fig. 6c). Within Alight there were 1 up / 0 down genes; within dark, 2 up / 1 down, the latter not in a Polycomb state. The Polycomb-down module (37 genes taken from the pooled Col-0 + WS OSD-218 run) trends down within Alight (median $\log_2\text{FC} -0.31$) but up within dark (+0.16). In the full 12-sample model (expression $\sim$ condition + light + condition:light), only 4 genes genome-wide show a significant interaction ($\text{padj} < 0.05$, $|\log_2\text{FC}| > 1$), and the median |interaction| is 0.50. Neither the redox module (23 genes; median |interaction| 1.26) nor the Polycomb-down module (37 genes; median 1.18) contains a significant interaction. The additive null is not explained by a large masked interaction. Taking the module from the Col-0-only OSD-218 run instead (58 genes; Table S67; Fig. 8b) gives the same picture: no gene with a significant interaction, median |interaction| 0.93, and module medians of -0.31 within Alight and $+0.11$ within dark. The redox-module and pooled-module rows reproduce Table S44 exactly (Table S68). But at 3 v 3 per stratum this is a bound on detectable interaction, not evidence of absence: OSD-120’s design cannot resolve the interaction on the Polycomb module.

3 Discussion

Three findings organise this work. First, **spaceflight repression is chromatin-state-biased**. Down-regulated genes concentrate in Polycomb-repressed, promoter-accessible euchromatin across datasets, tissues and state resolutions, while constitutive heterochromatin is inert. The bias is not

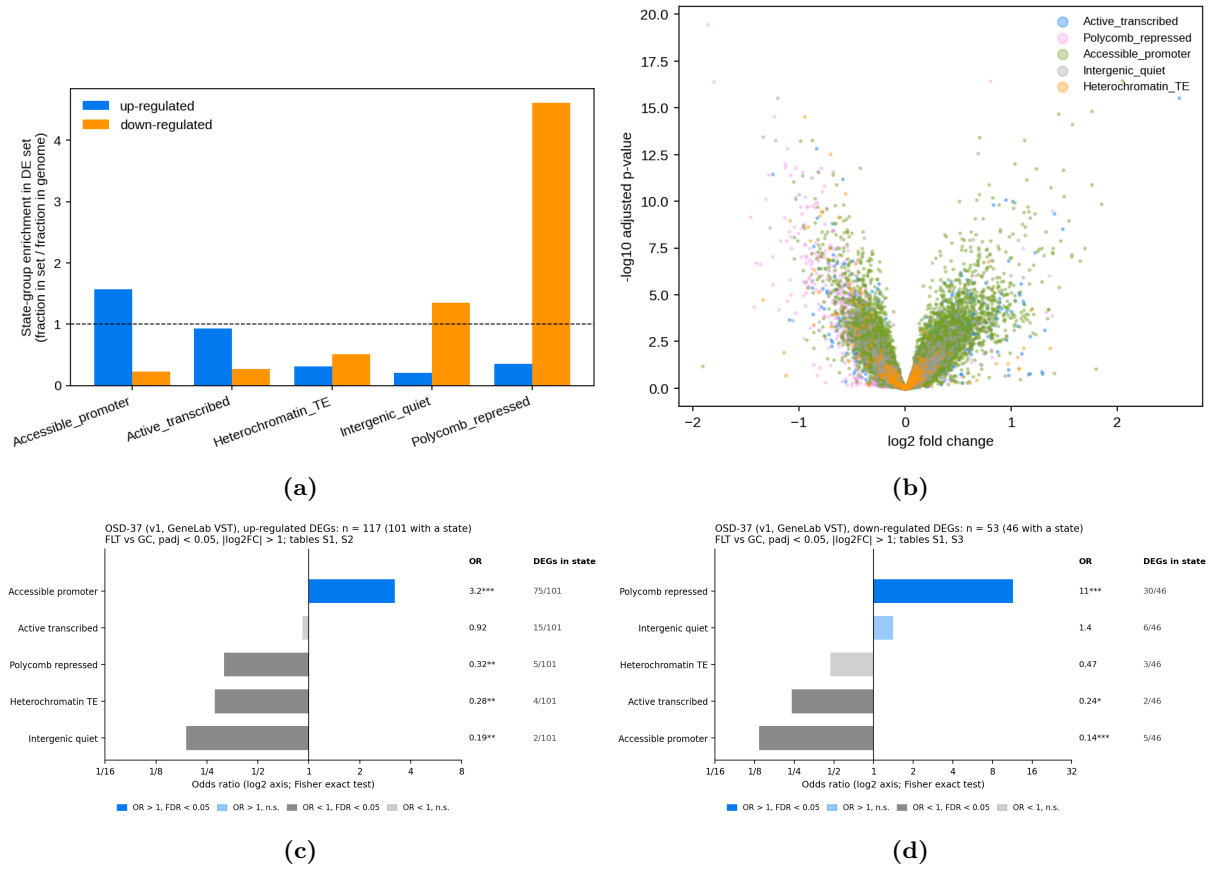
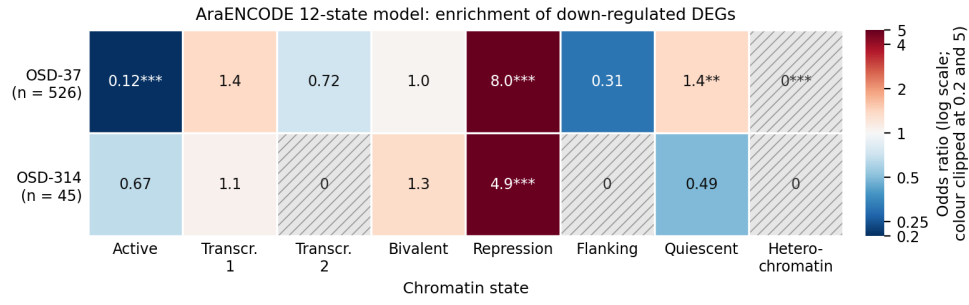


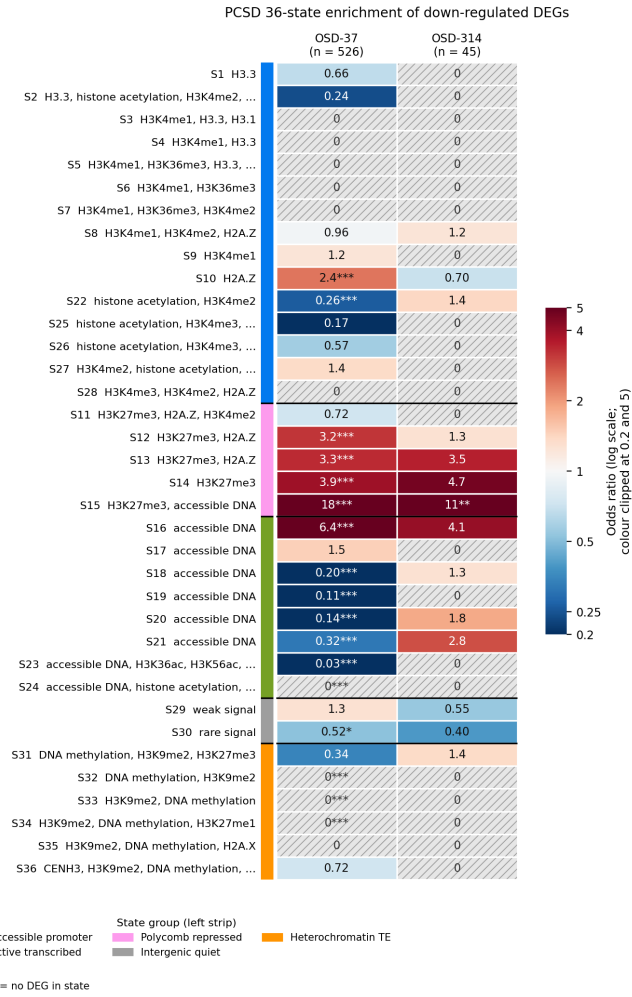
Figure 1: The decoder on OSD-37 (v1). (a) Chromatin-state distribution of genes. (b) Volcano plot coloured by chromatin-state group. (c, d) State-group enrichment of up- and down-regulated genes (Tables S2–S3).



0 = no DEG in state (OR 0)

Rows: datasets with ≥ 10 down-regulated DEGs ($\text{padj} < 0.05$, $|\log_2\text{FC}| > 1$). Not shown: OSD-120 (n=9), OSD-251 (n=1), OSD-346 (n=1). * FDR < 0.05 , ** < 0.01 , *** < 0.001 (BH). Source: sweep_enrichment12_long.csv (S11).

(a)



Columns: datasets with ≥ 10 down-regulated DEGs ($\text{padj} < 0.05$, $|\log_2\text{FC}| > 1$); not shown: OSD-120 (n=9), OSD-251 (n=1), OSD-346 (n=1). * FDR < 0.05 , ** < 0.01 , *** < 0.001 (BH). Mark lists shortened (...); full lists in the PCSD state table. Source: sweep_enrichment36_long.csv (S12).

(b)

Figure 2: OSDR-wide sweep: down-regulated genes concentrate in Polycomb states. (a) AraENCODE 12-state enrichment of down-regulated DEGs across datasets with DE (Table S11). (b) PCSD 36-state enrichment (Table S12); colour is clipped at odds 0.2 and 5. Only datasets with ≥ 10 DEGs are shown, and the OSD-314 column uses the published pooled contrast (Fig. 7). Up-regulated counterparts are Figs. S7–S8.

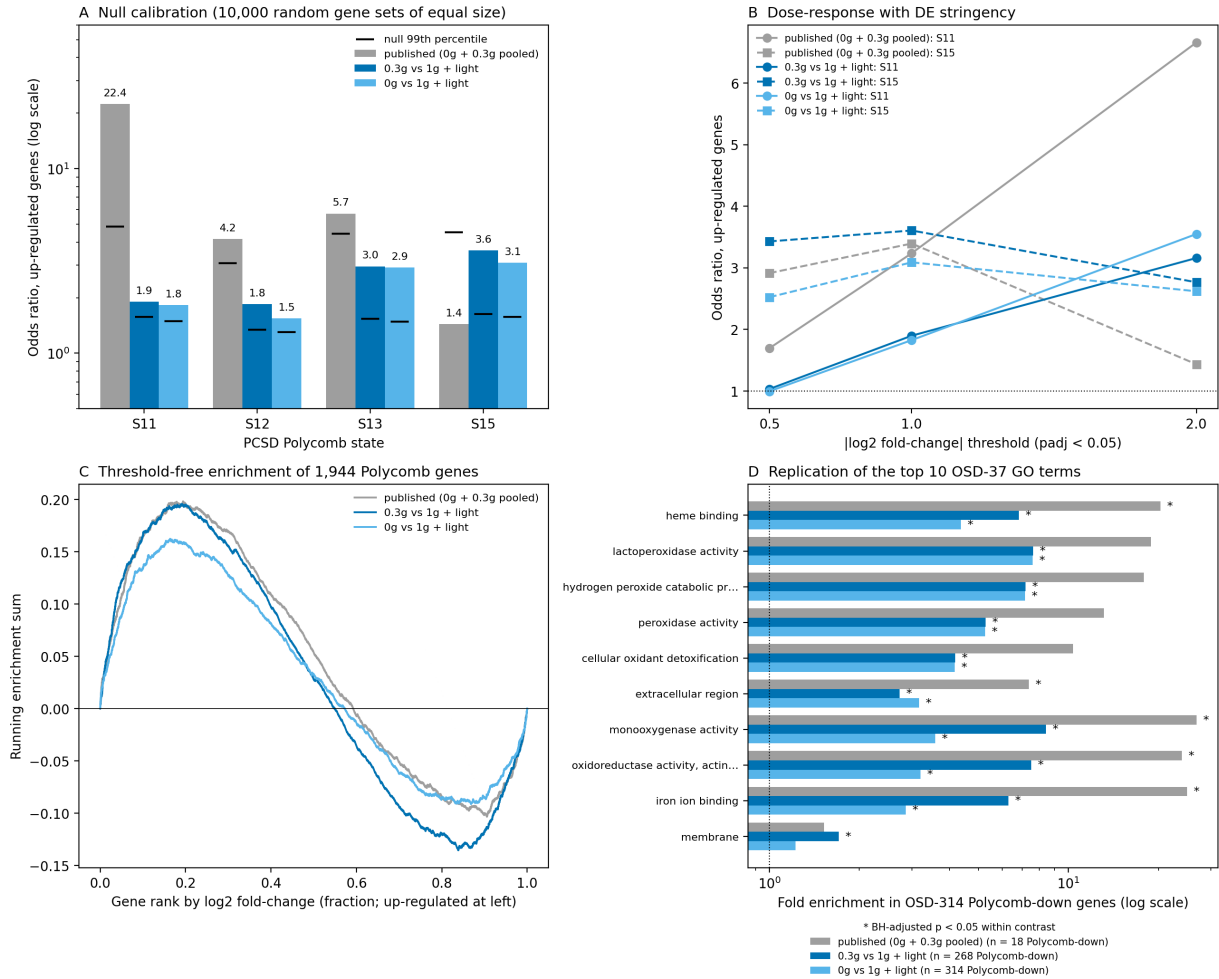


Figure 3: OSD-314 robustness diagnostics: published vs corrected contrasts. Grey: the published contrast, which pooled 0g and 0.3g against 1g. Dark blue: 0.3g vs 1g with light as a covariate (primary). Light blue: 0g vs 1g with light. (a) Odds ratio of up-regulated genes in each Polycomb state; bars mark the 99th percentile of 10,000 random gene sets of equal size (Table S58). (b) S11 and S15 odds across $|\log_2 \text{FC}|$ thresholds at padj < 0.05 (Table S60). (c) Running enrichment of 1,944 Polycomb genes along the fold-change ranking (Table S61). (d) Fold enrichment of the top 10 OSD-37 GO terms among OSD-314 Polycomb-down genes; * BH < 0.05 within contrast (Table S63). Generated by code/qc_osd314_figure.py. The original published-set diagnostics are Fig. S21.

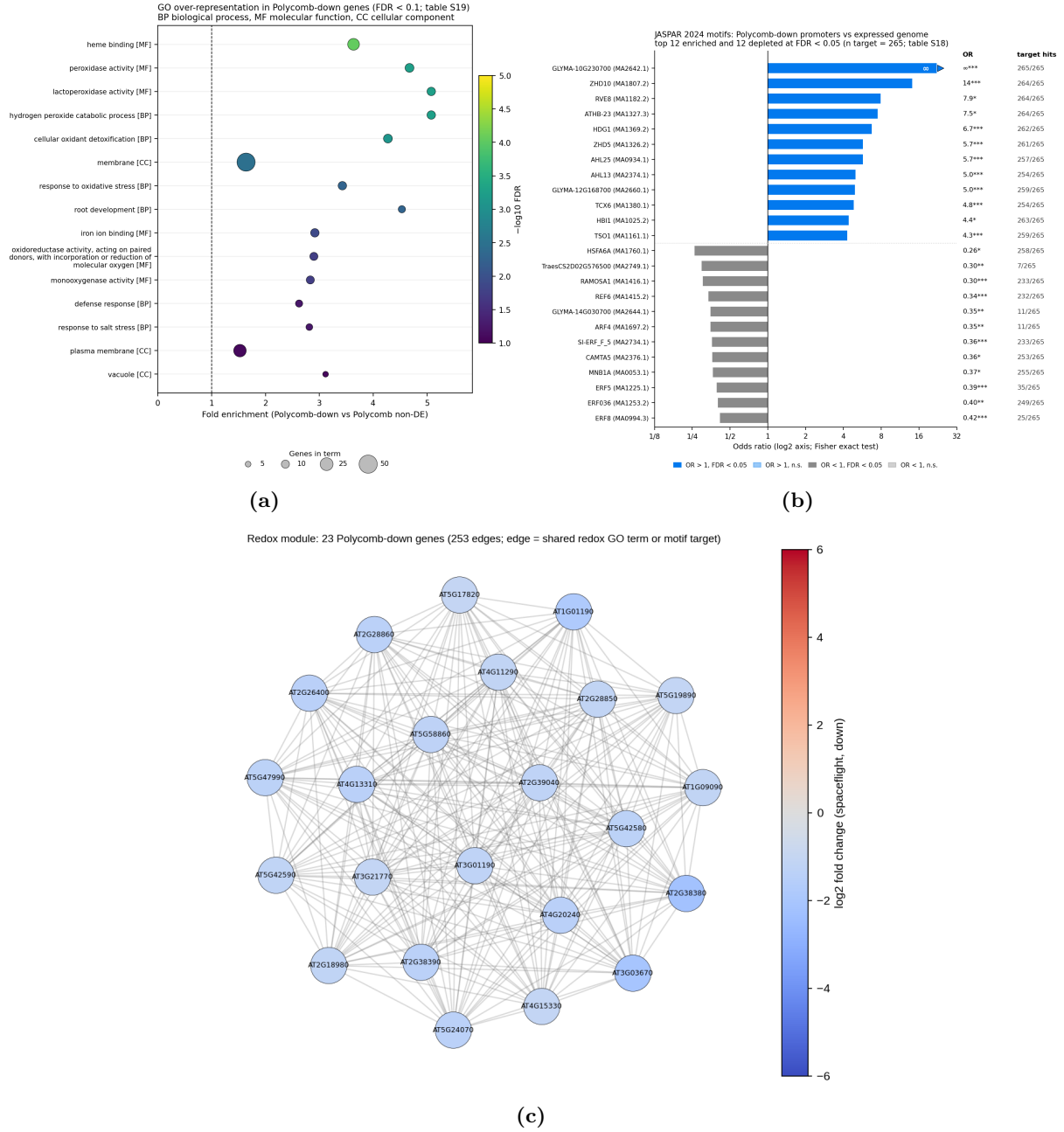


Figure 4: The repressed Polycomb class is a redox/oxidoreductase module. (a) GO over-representation against the genome and against the state-matched Polycomb control (Table S19). (b) JASPAR motif odds ratios in Polycomb-down promoters (Table S18). (c) The 23-gene redox module, connected by shared redox GO terms or shared top-motif targets. Node colour is the spaceflight log₂FC (Table D3).

expression-driven, since the learned layer fails honestly, and it is not explained by promoter GC content (state-matched controls).

Second, **the repressed class is a coherent redox/peroxidase module**. It is distinguishable within the Polycomb class by function (11 GO terms) but not by promoter motif architecture: the AHL/ZHD motifs track the state, not the response. This decouples “which genes are in Polycomb states” from “which Polycomb genes respond to spaceflight”, which is the central question the project reduces to.

Third, **the mechanism is not the canonical damage pathway**. SOG1, the strongest candidate Polycomb-recruiting TF with available ChIP-seq, does not bind the module’s promoters after DNA damage. Together with the methylation nulls, this leaves the driver of the state bias unidentified. Candidates include AHL/ZHD-family chromatin-associated factors (the motif layer), light/circadian machinery, and campaign-specific hardware effects (the OSD-624 null).

The power analysis turns the field’s inconsistent root results into a design requirement. Most root datasets in OSDR are underpowered for this test by a factor of two or more, and a stratified reanalysis cannot rescue an underpowered factorial design. The practical upshot is that the Polycomb-state bias is a robust, replicable feature of the *Arabidopsis* spaceflight transcriptome. The field needs a spaceflight H3K27me3 ChIP-seq or CUT&RUN assay with ≥ 8 replicates per group of a single genotype to convert the ground-state-decoded bias into a demonstrated dynamic mark change.

Limitations

- All chromatin states are **ground-state annotations**. The decoder tests enrichment of response genes in pre-existing states, not dynamic mark changes.
- OSD-193 and OSD-218 share a payload and lab. OSD-624’s null may reflect DE depth (36 down genes) or genuine campaign differences, and tissue is confounded with campaign.
- Motif and GO findings are correlative and discovery-only; AHL/ZHD binding has not been validated by DAP-seq or ChIP.
- Methylation tests are power-limited (3 replicates; 14–72 module genes in Ws) and ecotype-mismatched to the Col-0 discovery datasets.
- The SOG1 ChIP-seq is from bleomycin-treated, mixed Ler/Col seedlings at two timepoints. Binding under other damage regimes or in other tissues is not excluded.
- The power curve inherits OSD-218’s effect size and dispersion. Its Col-0-only version is limited to $k \leq 8$ by the 8 v 8 pool, so it cannot be extended to larger designs. The OSD-120 stratified strata (3 v 3) are underpowered, so the interaction result is a bound.
- OSD-406 and OSD-624 are suborbital (minutes of microgravity), and the EMCS partial-gravity datasets cannot answer the FLT/GC question as designed.
- The OSD-314 0 g arm may be batch-confounded (its sample identifiers differ from the 1 g and 0.3 g samples), so the 0.3 g contrast is treated as primary. The original OSD-314 diagnostics (Tables S13–S17, S21) describe the pooled contrast and are superseded by Tables S58–S63.

4 Methods

Pipeline

Python 3.11. The core module `chromdecode.py` provides: ID harmonisation (`clean_agi`); counts loading; DE (`differential_expression_any`: linear models on $\log_2(\text{counts})$ with condition + covariates, BH-FDR); coordinate mapping (TAIR10 [23], TSS-anchored gene body + 2 kb); state

assignment (PCSD 36-state segments, AraENCODE 12-state seedling track, in which 8 of the 12 state labels occur at gene TSSs; PlantCADB ACR overlap); enrichment (Fisher exact, BH-FDR); and an expression-only fallback decoder. A check script (`test_chromdecode.py`) runs 21 checks on loci with known annotations.

Data

PCSD (36 ChromHMM states, 290,553 segments) [4]; AraENCODE (12 states, TPM matrix, gene annotation) [6]; PlantCADB ACRs [8]; NASA OSDR metadata API and GeneLab-processed counts [1, 2]; JASPAR 2024 plant PFMs [12]; TAIR GO annotations and `go.obo` [11]; and the TAIR10 genome. The source data are not redistributed. Accessions and expected paths are listed in `DATA_SOURCES.md`.

ChIP-seq reanalysis

HISAT2 2.2.1 on an index built from TAIR10 (single-end, `-no-spliced-alignment`); samtools sort/index; MACS2 2.2.9.1 `callpeak` (`-f BAM -g 1.2e8 -q 0.01`; IP vs matched-timepoint input; wild-type-IP specificity check). Promoter overlap uses strand-aware 2 kb TSS intervals, tested with Fisher’s exact test against an expressed non-DE background and BH-FDR. Occupancy is samtools per-100-bp bin counts scaled to RPM. The commands are in `code/hpc/sog1_alignment.md`.

Metadata forensics

NCBI E-utilities (`esearch + esummary, db=biosample`) for GSM/SAMN sample IDs; titles were parsed into condition, age, genotype and replicate. Median-ratio size-factor scaling was used where only unnormalised counts exist.

Power and interaction analyses

Empirical power: the full pipeline was re-run on $k \in \{2, 3, 4, 6, 8, 12, 16\}$ replicates per group subsampled from OSD-218, with 20 draws per k , with and without two random binary nuisance covariates mimicking OSD-120’s light $\times$ day design. Analytic power: simulated Fisher tests at odds 3.1 and 7.0 against a background fraction of 0.141. Interaction: gene-wise linear model $\text{expression} \sim \text{condition} + \text{light} + \text{condition}:\text{light}$ on the 12 OSD-120 Col-0 WT samples, with BH-FDR across genes.

Pre-submission audit and corrections

Before submission, every dataset’s platform, genotypes and design were re-read from its OSDR study record and NCBI BioSample titles. Every sweep result was then regenerated from the public counts with the pinned environment (`code/qc_sweep_rerun.py`, `code/qc_genotype_rerun.py`). The re-run reproduced the archived counts and odds ratios exactly (Table S50; S48, first two rows), and it found four problems, all corrected in the text above. (i) OSD-314’s Mars-gravity samples (“03g”) were not parsed, and the contrast coding counted them as treatment (Table S53). (ii) The BioSample parser did not recognise “WS”, so OSD-218’s 16 WS samples entered a contrast described as Col-0. (iii) OSD-406 was described as Col-0, but it holds Col-0, WS and *sku5* from two rockets. (iv) OSD-624 was described as an orbital dataset; it is a Virgin Galactic suborbital flight. Two archive-level problems were also fixed. The 5-group sweep table exported from the analysis workspace predated the $\log_2$ correction, and it has been replaced by a regenerated table (Table S51). The significance stars in the original 5-group heatmap were misaligned by a column-order bug in `aggregate_sweep.py`. The OSD-314 diagnostics and GO replication were re-run on the corrected contrasts (`code/qc_osd314_diagnostics.py`, which first reproduces the published tables exactly; Table S64). The power subsampling and the OSD-120 module bound

were re-run with OSD-218 Col-0 only (`code/qc_col0_power_bound.py`, which first reproduces Table S42–Table S44 exactly; Table S68).

Statistics

Fisher exact (one-sided where enrichment is directional); BH-FDR within test families; hypergeometric GO over-representation; permutation-based null calibration and GSEA (1,000–10,000 draws); genomic-block CV for the learned layer. Thresholds are $\text{padj} < 0.05$ and $|\log_2 \text{FC}| > 1$ unless stated. All results are discovery-only.

Data availability

All result tables are in `supplementary_tables/` (Tables S1–S68, indexed in its `README.md`), with small derived data in `data/derived/` (Tables D1–D7). All 52 figures (SVG + PNG) are in `figures/`. The complete per-version project record is `reports/report_chromdecode_at.md`. Source data are public and listed in `DATA_SOURCES.md`. Repository: <https://github.com/dr-richard-barker/ChromDecode-At>. Archived release: Zenodo DOI to be assigned on first release.

Code availability

All 33 analysis scripts are in `code/` (MIT licence). `chromdecode.py` is the core module.

Competing interests

The author declares no competing interests.

References

- [1] Samrawit G Gebre, Ryan T Scott, Amanda M Saravia-Butler, Danielle K Lopez, Lauren M Sanders, and Sylvain V Costes. NASA open science data repository: open science for life in space. *Nucleic Acids Research*, 53(D1):D1697–D1710, 2025. doi: 10.1093/nar/gkae1116.
- [2] Daniel C Berrios, Jonathan Galazka, Kirill Grigorev, Samrawit Gebre, and Sylvain V Costes. NASA GeneLab: interfaces for the exploration of space omics data. *Nucleic Acids Research*, 49(D1):D1515–D1522, 2021. doi: 10.1093/nar/gkaa887.
- [3] Anna-Lisa Paul, Natasha J. Sng, Agata K. Zupanska, Aparna Krishnamurthy, Eric R. Schultz, and Robert J. Ferl. Genetic dissection of the Arabidopsis spaceflight transcriptome: Are some responses dispensable for the physiological adaptation of plants to spaceflight? *PLOS ONE*, 12(6):e0180186, 2017. doi: 10.1371/journal.pone.0180186.
- [4] Yue Liu, Tian Tian, Kang Zhang, Qi You, Hengyu Yan, Nannan Zhao, Xin Yi, Wenying Xu, and Zhen Su. PCSD: a plant chromatin state database. *Nucleic Acids Research*, 46(D1):D1157–D1167, 2018. doi: 10.1093/nar/gkx919.
- [5] Jason Ernst and Manolis Kellis. ChromHMM: automating chromatin-state discovery and characterization. *Nature Methods*, 9(3):215–216, 2012. doi: 10.1038/nmeth.1906.
- [6] Zhenji Wang, Minghao Liu, Fuming Lai, Qiangqiang Fu, Liang Xie, Yaping Fang, Qiangwei Zhou, and Guoliang Li. AraENCODE: A comprehensive epigenomic database of Arabidopsis thaliana. *Molecular Plant*, 16(7):1113–1116, 2023. doi: 10.1016/j.molp.2023.06.005.

- [7] Yoav Benjamini and Yosef Hochberg. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.
- [8] Ke Ding, Shanwen Sun, Yang Luo, Chaoyue Long, Jingwen Zhai, Yixiao Zhai, and Guohua Wang. PlantCADB: A Comprehensive Plant Chromatin Accessibility Database. *Genomics, Proteomics & Bioinformatics*, 21(2):311–323, 2023. doi: 10.1016/j.gpb.2022.10.005.
- [9] Won-Gyu Choi, Richard J. Barker, Su-Hwa Kim, Sarah J. Swanson, and Simon Gilroy. Variation in the transcriptome of different ecotypes of *Arabidopsis thaliana* reveals signatures of oxidative stress in plant responses to spaceflight. *American Journal of Botany*, 106(1): 123–136, 2019. doi: 10.1002/ajb2.1223.
- [10] Alicia Villacampa, Malgorzata Ciska, Aránzazu Manzano, Joshua P. Vandenbrink, John Z. Kiss, Raúl Herranz, and F. Javier Medina. From Spaceflight to Mars g-Levels: Adaptive Response of *A. Thaliana* Seedlings in a Reduced Gravity Environment Is Enhanced by Red-Light Photostimulation. *International Journal of Molecular Sciences*, 22(2):899, 2021. doi: 10.3390/ijms22020899.
- [11] The Gene Ontology Consortium. The Gene Ontology knowledgebase in 2023. *Genetics*, 224(1):iyad031, 2023. doi: 10.1093/genetics/iyad031.
- [12] Ieva Rauluseviciute, Rafael Riudavets-Puig, Romain Blanc-Mathieu, Jaime A Castro-Mondragon, Katalin Ferenc, Vipin Kumar, Roza Berhanu Lemma, Jérémy Lucas, Jeanne Chèneby, Damir Baranasic, Aziz Khan, Oriol Fornes, Sveinung Gundersen, Morten Johansen, Eivind Hovig, Boris Lenhard, Albin Sandelin, Wyeth W Wasserman, François Parcy, and Anthony Mathelier. JASPAR 2024: 20th anniversary of the open-access database of transcription factor binding profiles. *Nucleic Acids Research*, 52(D1):D174–D182, 2024. doi: 10.1093/nar/gkad1059.
- [13] Mingqi Zhou, Natasha J. Sng, Collin E. LeFrois, Anna-Lisa Paul, and Robert J. Ferl. Epigenomics in an extraterrestrial environment: organ-specific alteration of DNA methylation and gene expression elicited by spaceflight in *Arabidopsis thaliana*. *BMC Genomics*, 20(1): 205, 2019. doi: 10.1186/s12864-019-5554-z.
- [14] Yuanxin Xi and Wei Li. BSMAP: whole genome bisulfite sequence MAPping program. *BMC Bioinformatics*, 10(1):232, 2009. doi: 10.1186/1471-2105-10-232.
- [15] Deqiang Sun, Yuanxin Xi, Benjamin Rodriguez, Hyun Jung Park, Pan Tong, Mira Meong, Margaret A Goodell, and Wei Li. MOABS: model based analysis of bisulfite sequencing data. *Genome Biology*, 15(2):R38, 2014. doi: 10.1186/gb-2014-15-2-r38.
- [16] Kaoru Yoshiyama, Phillip A. Conklin, Neil D. Huefner, and Anne B. Britt. Suppressor of gamma response 1 (SOG1) encodes a putative transcription factor governing multiple responses to DNA damage. *Proceedings of the National Academy of Sciences*, 106(31): 12843–12848, 2009. doi: 10.1073/pnas.0810304106.
- [17] Clara Bourbousse, Neeraja Vegesna, and Julie A. Law. SOG1 activator and MYB3R repressors regulate a complex DNA damage network in *Arabidopsis*. *Proceedings of the National Academy of Sciences*, 115(52):E12453–E12462, 2018. doi: 10.1073/pnas.1810582115.
- [18] Daehwan Kim, Joseph M. Paggi, Chanhee Park, Christopher Bennett, and Steven L. Salzberg. Graph-based genome alignment and genotyping with HISAT2 and HISAT-genotype. *Nature Biotechnology*, 37(8):907–915, 2019. doi: 10.1038/s41587-019-0201-4.

- 440 [19] Yong Zhang, Tao Liu, Clifford A Meyer, Jérôme Eeckhoutte, David S Johnson, Bradley E
441 Bernstein, Chad Nusbaum, Richard M Myers, Myles Brown, Wei Li, and X Shirley Liu.
442 Model-based Analysis of ChIP-Seq (MACS). *Genome Biology*, 9(9):R137, 2008. doi: 10.1186/
443 gb-2008-9-9-r137.
- 444 [20] Brandon Califar, Natasha J. Sng, Agata Zupanska, Anna-Lisa Paul, and Robert J. Ferl.
445 Root Skewing-Associated Genes Impact the Spaceflight Response of *Arabidopsis thaliana*.
446 *Frontiers in Plant Science*, 11:239, 2020. doi: 10.3389/fpls.2020.00239.
- 447 [21] Robert J. Ferl, Mingqi Zhou, Hunter F. Strickland, Natasha J. Haveman, Jordan B. Callaham,
448 Sirisha Bandla, Daniel Ambriz, and Anna-Lisa Paul. Transcriptomic dynamics in the transition
449 from ground to space are revealed by Virgin Galactic human-tended suborbital spaceflight.
450 *npj Microgravity*, 9(1):95, 2023. doi: 10.1038/s41526-023-00340-w.
- 451 [22] Brandon Califar, Agata Zupanska, Jordan A. Callaham, Matthew T. Bamsey, Thomas
452 Graham, Anna-Lisa Paul, and Robert J. Ferl. Shared Metabolic Remodeling Processes
453 Characterize the Transcriptome of *Arabidopsis thaliana* within Various Suborbital Flight En-
454 vironments. *Gravitational and Space Research*, 9(1):13–29, 2021. doi: 10.2478/gsr-2021-0002.
- 455 [23] Tanya Z. Berardini, Leonore Reiser, Donghui Li, Yarik Mezheritsky, Robert Muller, Emily
456 Strait, and Eva Huala. The *arabidopsis* information resource: Making and mining the
457 “gold standard” annotated reference plant genome. *Genesis*, 53(8):474–485, 2015. doi:
458 10.1002/dvg.22877.

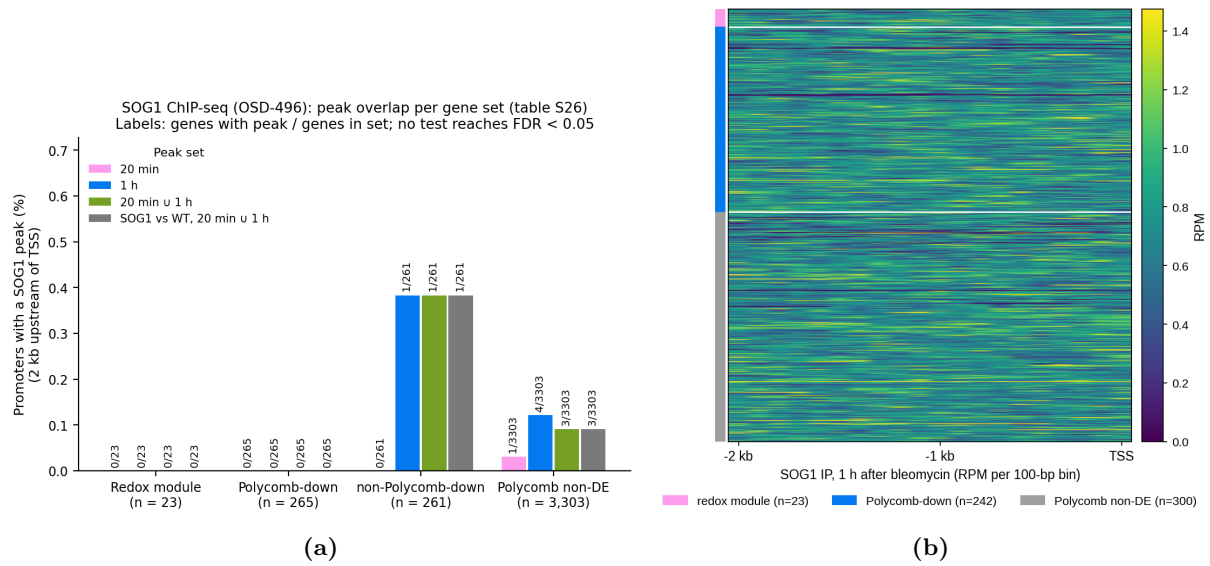


Figure 5: SOG1 does not bind redox-module or Polycomb-down promoters after DNA damage (OSD-496). (a) Fraction of promoters with a SOG1 peak, per gene set and timepoint (Table S26). (b) 1-h SOG1 promoter occupancy (RPM per 100-bp bin) by gene set.

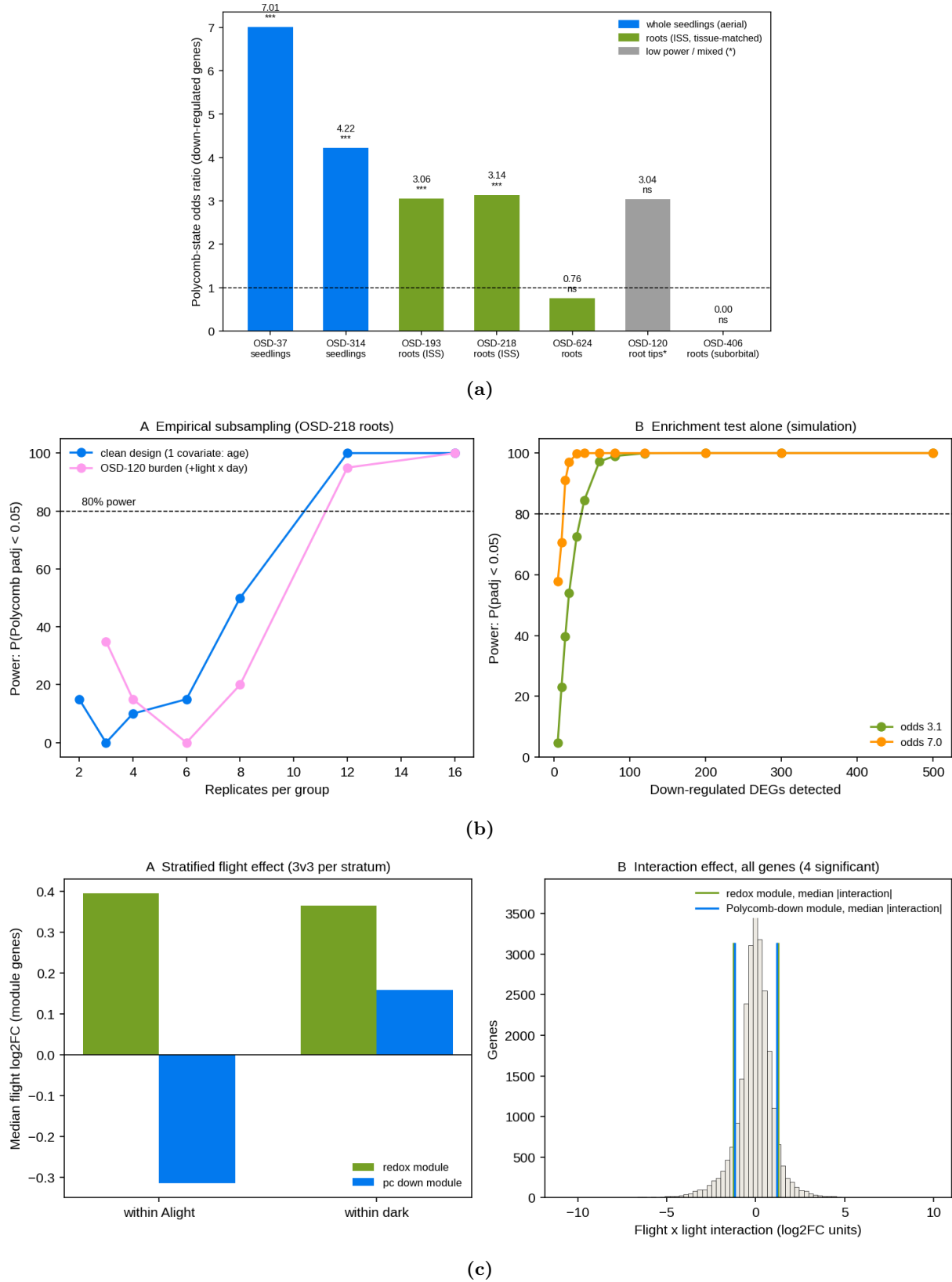


Figure 6: Root replication, power and the OSD-120 interaction bound. (a) Polycomb-state odds ratio of down-regulated genes per dataset, as computed in v8 (Tables S29, S32, S34, S37, S39, S41). The OSD-314, OSD-218 and OSD-406 bars use the v8 contrasts; Fig. 7 gives their corrected values. OSD-624 and OSD-406 are suborbital flights. (b) Power by replicates per group (empirical OSD-218 subsampling) and by number of down-regulated DEGs (simulation; Tables S42–S43). (c) OSD-120 Col-0 WT flight effect within each light stratum and the gene-level condition $\times$ light interaction (Tables S44–S47).

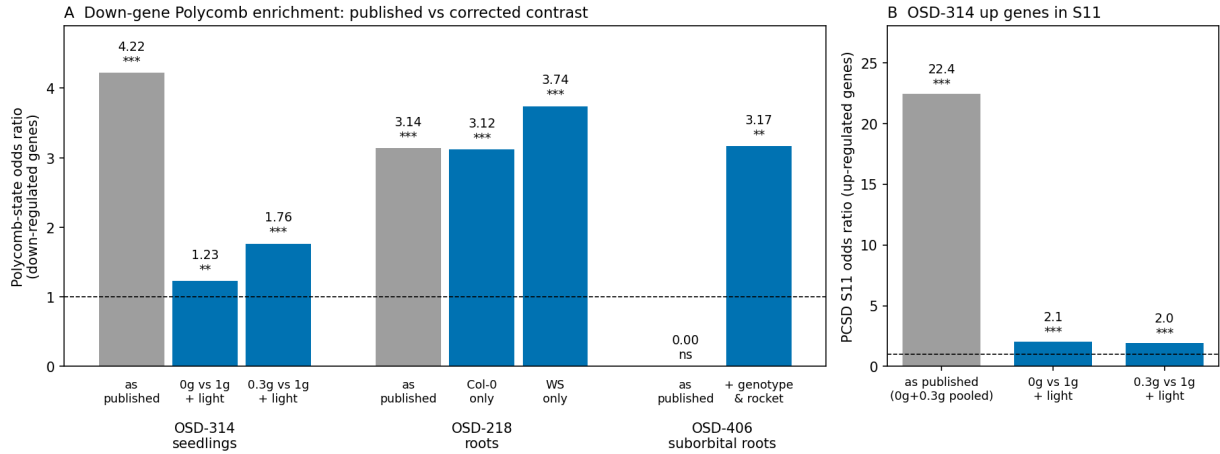


Figure 7: What the pre-submission audit changed. (a) Down-regulated-gene Polycomb (5-group) odds ratio under the published contrast (grey) and the corrected contrast (blue). OSD-314: the published contrast pooled 0g and Mars 0.3g against 1g, and the corrected contrasts separate the arms and add light as a covariate (Table S54). OSD-218: the published run pooled Col-0 and WS, and the corrected runs split them by genotype (Table S48). OSD-406: the published run pooled three genotypes and two rockets without covariates (Table S48). (b) OSD-314 up-regulated genes in PCSD state S11, published vs corrected. Stars: BH-FDR < 0.05 (*), < 0.01 (**), < 0.001 (***). Generated by `code/qc_figure.py`.

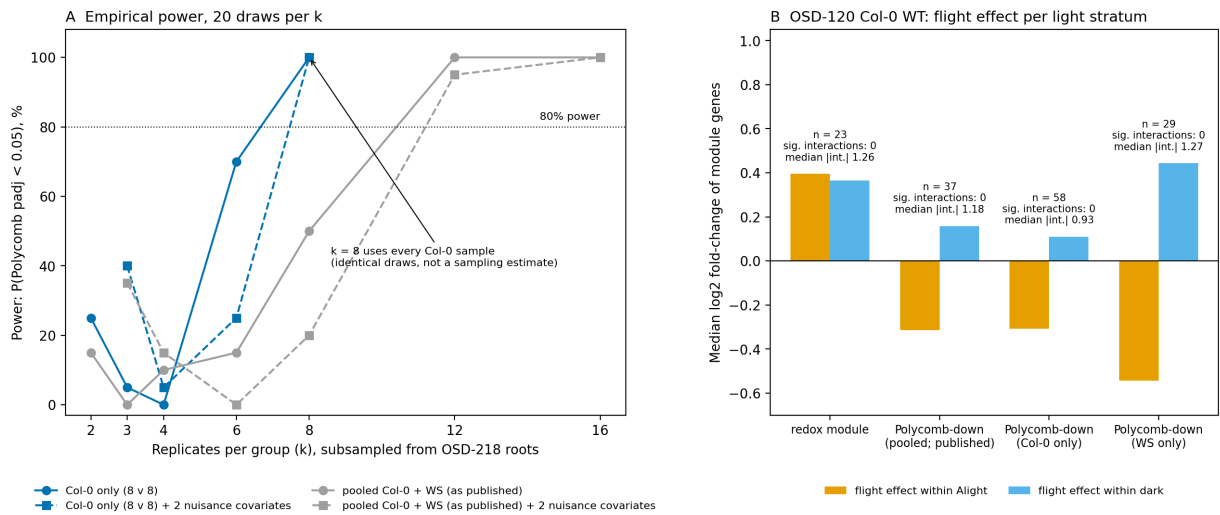


Figure 8: Power and the OSD-120 interaction bound with OSD-218 Col-0 only. (a) Empirical power of the Polycomb enrichment test by replicates per group, from 20 subsampled draws per k . Grey: the published pooled Col-0 + WS pool (16 v 16). Blue: Col-0 only (8 v 8). Dashed: two random nuisance covariates mimicking OSD-120's design. At $k = 8$ the Col-0 draws are identical (Table S43, Table S65). (b) Median flight $\log_2$ FC of module genes within each OSD-120 light stratum, for the redox module and for Polycomb-down modules taken from pooled, Col-0-only and WS-only OSD-218 runs. Annotations give module size, number of genes with a significant flight $\times$ light interaction, and median |interaction| (Table S67). Generated by `code/qc_col0_figure.py`.